

MOTIB ALANAZI

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EDUCATION

University of Utah | Salt Lake City, Utah, USA

B.S. Chemical Engineering - In Progress

Relevant Coursework: Chemical Engineering Design & Innovation; Thermodynamics I; Fundamentals of Process Engineering; Numerical Methods; Statistics for Chemical Engineers; General Chemistry Laboratory I & II

ENGINEERING PROJECTS

AICHE Chem-E-Car - "Sparky"

1st Place - Regional AIChE Chem-E-Car Competition

- Co-developed and tested a chemically powered vehicle using a custom alkaline zinc battery with 2 M NaOH electrolyte, DC motor propulsion, and a reaction-triggered stopping system.
- Integrated a **fuchsin/NaOH chemical clock** with photometric sensing and Arduino control to trigger motor shutdown; supported chemical preparation, testing, troubleshooting, and safe handling.

Portable Optical Spectroscopy Device

Arduino Nano | Python | Optical Sensing | Circuit Assembly | CAD / 3D Printing

- Designed and built a portable optical measurement device with a light sensor, Arduino Nano, custom circuitry, and a 3D-printed light-controlled enclosure.
- Developed the measurement workflow, including **sensor calibration**, data acquisition, Python-based processing, and chemistry-based calculations to interpret optical measurements.

Conductive PEDOT:PSS Hydrogel - Capstone Research Support

- Supported hydrogel research by spin-coating **SU-8 2000** onto glass, acquiring microscopy images, and quantifying neurite growth using NeuronJ; exported and organized measurements in Excel for analysis.

SoundBoy - Real-Time Audio Processing System

Swift | Python | Core Audio | PyTorch

- Engineered a real-time multi-stream audio system with modular processing, bounded buffering, and native hardware audio acquisition.
- Integrated local machine-learning models through a versioned interprocess protocol; implemented overload detection, device-change recovery, and **fail-safe output behavior**. Measured approximately 165 ms p95 processing time per 500 ms conversion interval.

TECHNICAL EXPERIENCE

Certified Apple Technician / Student Supervisor

University of Utah Campus Store - Apple Authorized Service Provider | 2022-2024

- Completed **300+ Apple device repairs** with a **98% reported success rate**, diagnosing hardware and software faults through structured troubleshooting.
- Coordinated repair workflow, GSX documentation, device logistics, parts requirements, and technical priorities while following service, safety, and compliance procedures.
- Trained and onboarded technicians in diagnostics, repair procedures, and documentation standards; helped **coordinate daily technical operations** across technicians and management.

TECHNICAL SKILLS

Chemical / Laboratory: Process engineering; experimental & numerical data analysis; chemical handling; laboratory safety; SDS interpretation; spin coating

Instrumentation / Tools: Arduino Nano; optical sensing; photometry; sensor calibration; circuit assembly; microscopy; data acquisition; CAD; 3D printing; Excel; NeuronJ

Programming: Python; Swift; iOS application development; Core Audio; PyTorch; real-time systems; interprocess communication

Languages: Arabic - Native | English - Fluent | Spanish - Basic Conversational

AWARDS, MEMBERSHIPS & TRAINING

Gold Medal - KSU Math Olympics | Qualified for National Competition

American Institute of Chemical Engineers (AIChE) - Student Member | AIChE SDS Training

Fundamentals of Sustainable Energy Technologies - Saudi Nuclear Society